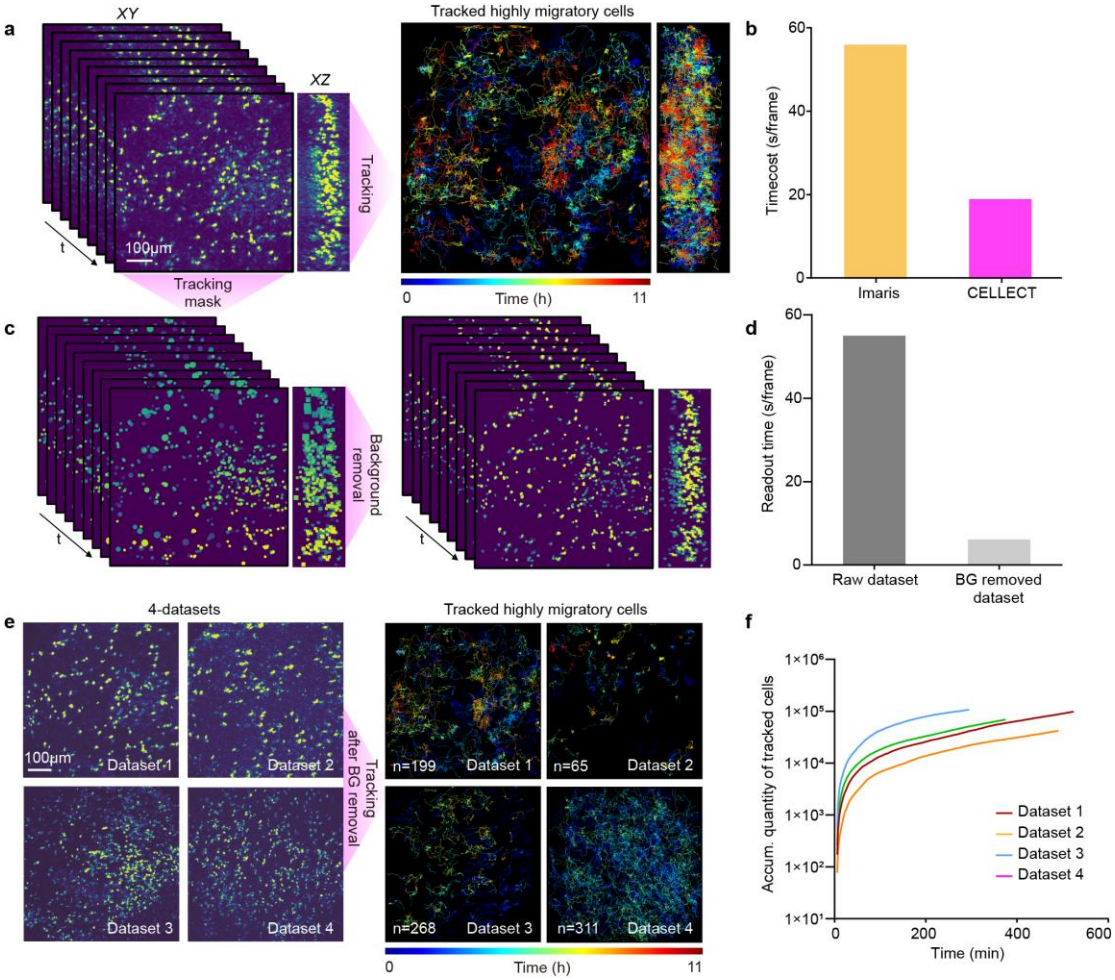


Supplementary Figure 18 | Tracking performance of CELLECT on terabyte-level imaging datasets of B-cell dynamics. a, Example MIPs from the data sequence and corresponding cell tracking results. **b**, Comparison of the running speed between CELLECT and Imaris. **c**, Background removal based on our tracking results, which reduces data storage size and subsequent reading time. **d**, Comparison of reading times after background removal. **e**, Display of long-distance migration trajectories across all 4 datasets. **f**, Integral representation of cell counts in the 4 datasets. All data were acquired using 2pSAM imaging of B-cell dynamics in mouse lymph nodes. These datasets are distinct from the one shown in **Fig. 3**, but were collected under identical imaging conditions and protocols. The total data volume exceeds 4 TB. CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset. Scale bars: 100 μm.